Projection-onto-HP-opposite axis vs distance from HP
Per-subject smoothed curves + 1D cluster-based permutation

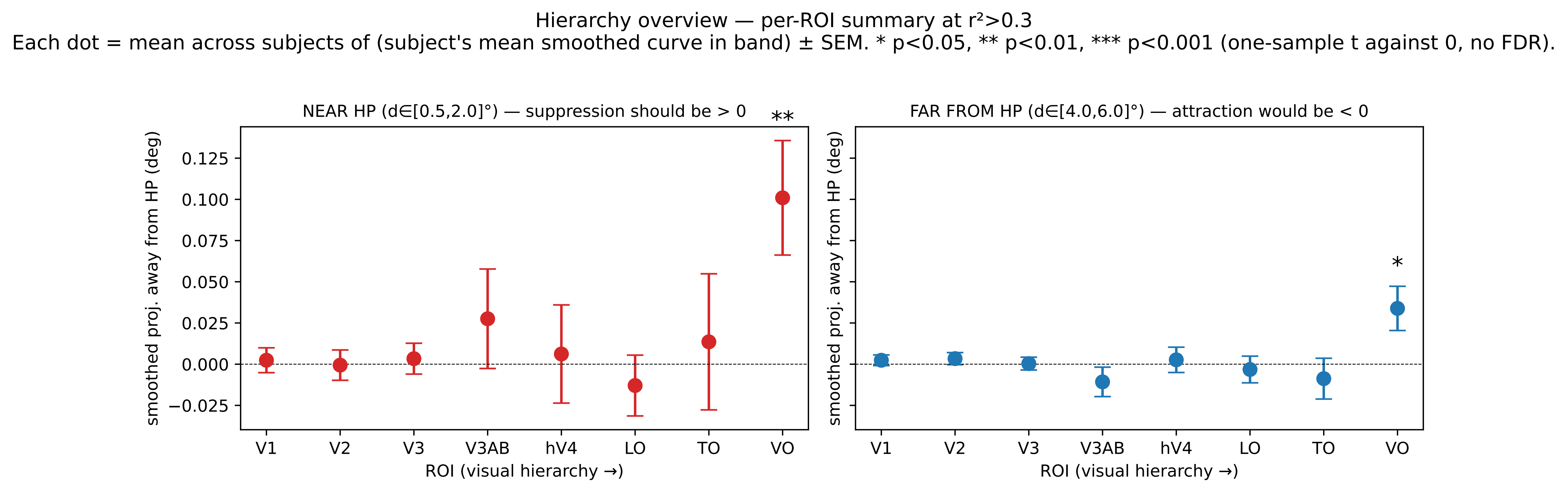
Inclusion: aperture filter (drop voxels with > 50% of any conditionwise PRF outside aperture).
Plus per-panel r^2_mean_model > threshold (varied across columns).

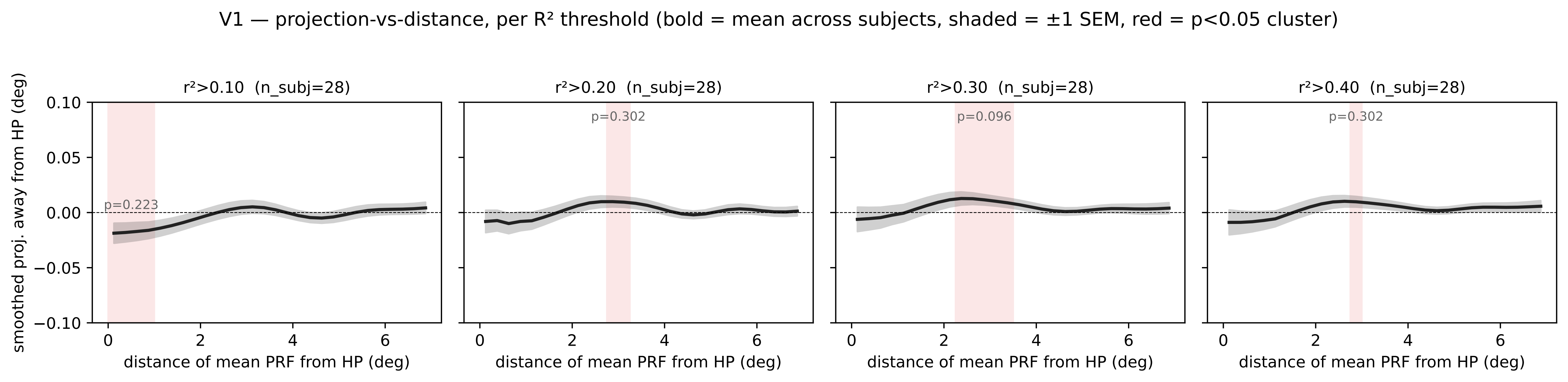
Pipeline:

- Per (subject, ROI, distance bin): mean over voxels of projection_away_from_HP = −dy_rotated, where rotated frame puts HP at canonical (0, +4°).
- Per subject: Gaussian-smooth across bins (σ = 1.0 bins).
- Group-level: BOLD LINE = mean across subjects of smoothed curves
  SHADED BAND = ±1 SEM across subjects.
- At each bin: one-sample t against 0. Bins with <60% subjects are skipped.
- Cluster: consecutive bins with |t| > 2.0.
  Cluster stat = Σ t inside.
- Permutation null: each subject's smoothed curve sign-flipped with p=0.5,
  300 permutations, max-cluster-stat distribution.
- Cluster significant if observed |stat| > 95th percentile of null.

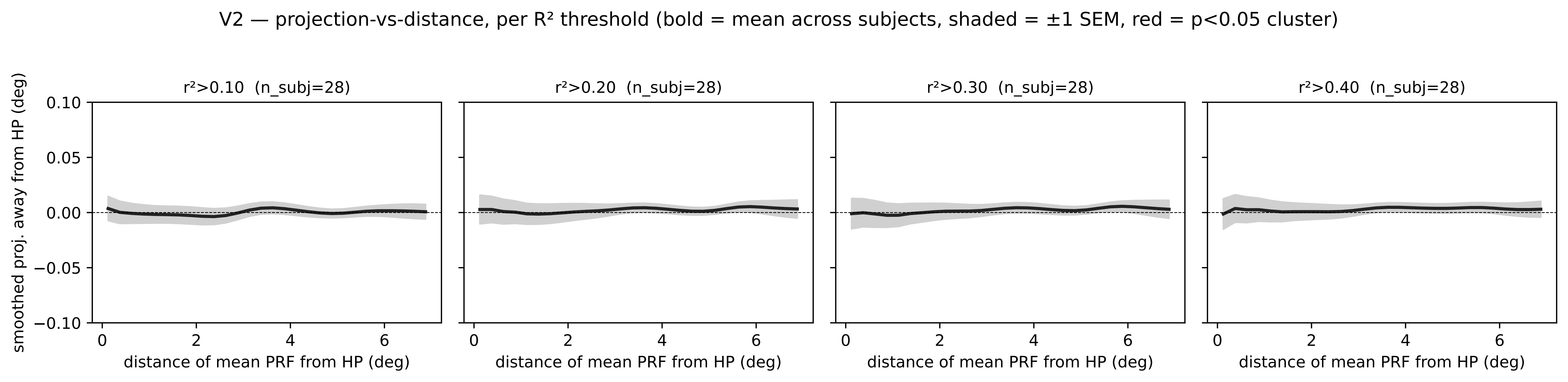
Bin width: 0.25°. |proj| clip: ±1.5°. y-axis: ±0.1°.

R^2 thresholds compared: [0.1, 0.2, 0.3, 0.4]
ROIs: ['V1', 'V2', 'V3', 'V3AB', 'hV4', 'LO', 'TO', 'VO']

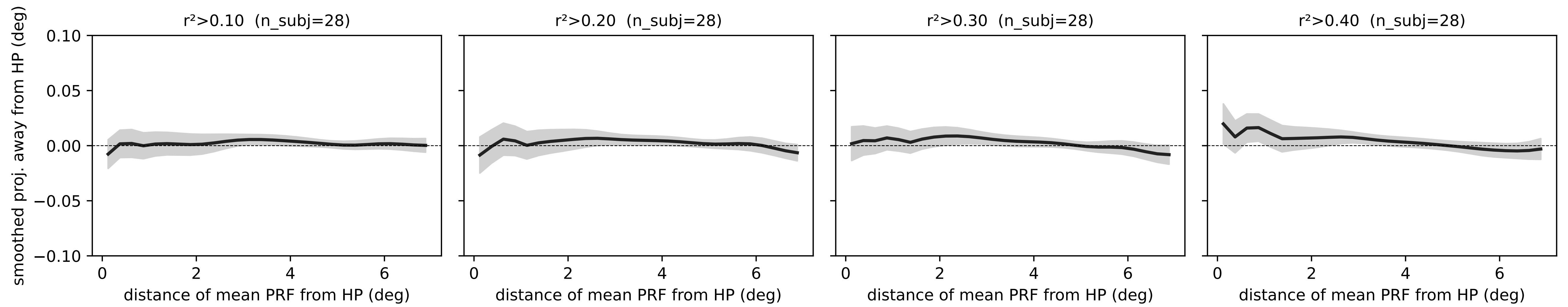

Hierarchy overview — per-ROI summary at r²>0.3
Each dot = mean across subjects of (subject's mean smoothed curve in band) ± SEM. * p<0.05, ** p<0.01, *** p<0.001 (one-sample t against 0, no FDR).
NEAR HP (d∈[0.5,2.0]°) — suppression should be > 0
FAR FROM HP (d∈[4.0,6.0]°) — attraction would be < 0
smoothed proj. away from HP (deg)
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
**
*
V1
V2
V3
V3AB
hV4
LO
TO
VO
ROI (visual hierarchy →)

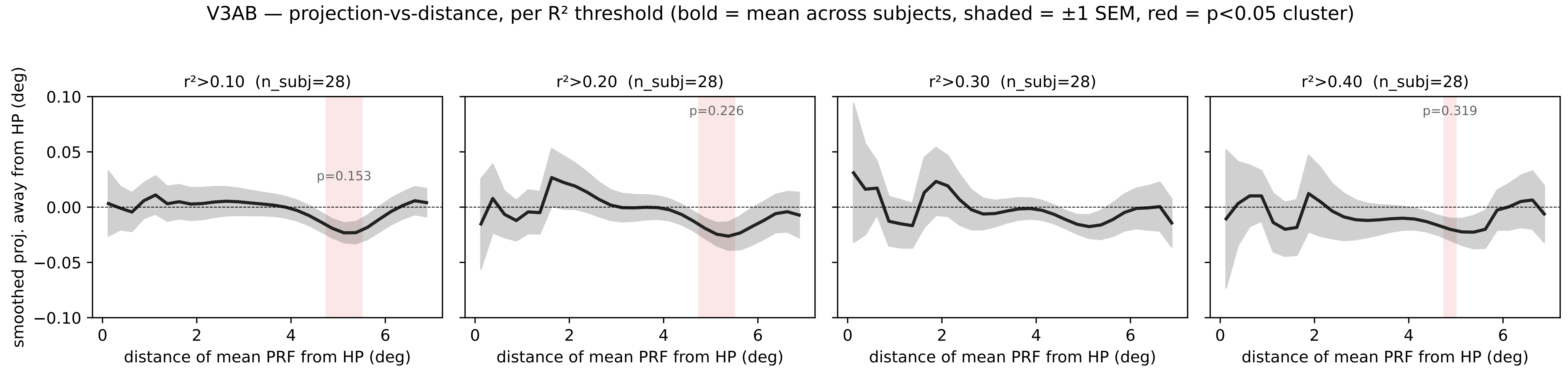


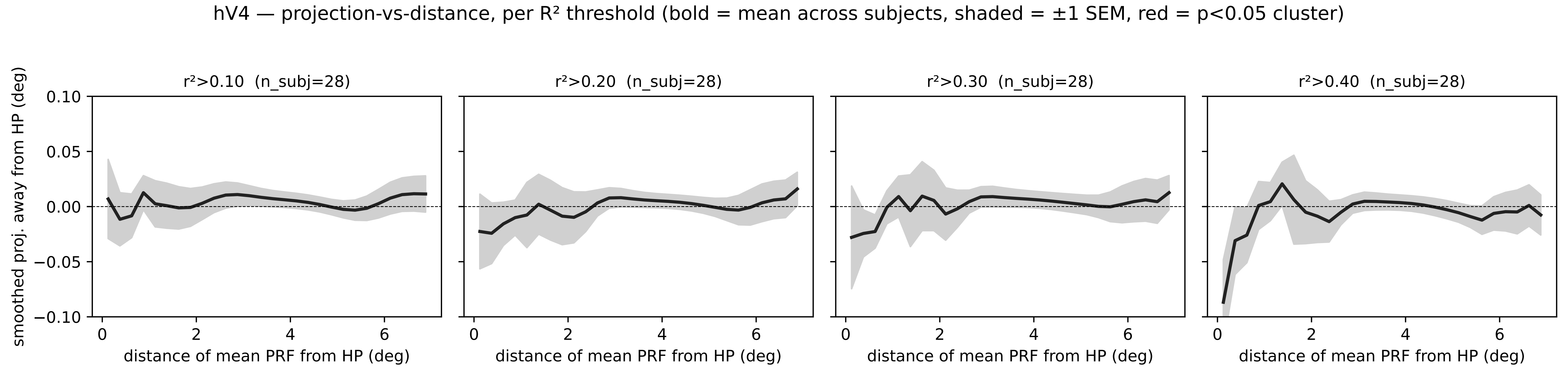
V1 — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=28)
r²>0.30 (n_subj=28)
r²>0.40 (n_subj=28)
p=0.223
p=0.302
p=0.096
p=0.302
smoothed proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6


V2 — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=28)
r²>0.30 (n_subj=28)
r²>0.40 (n_subj=28)
smoothed proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6

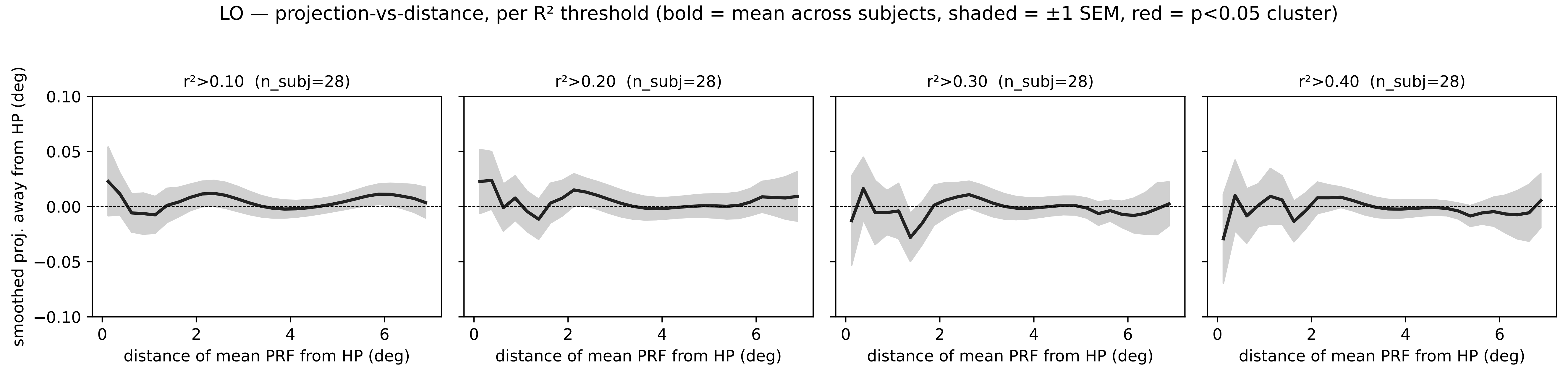
V3 — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)



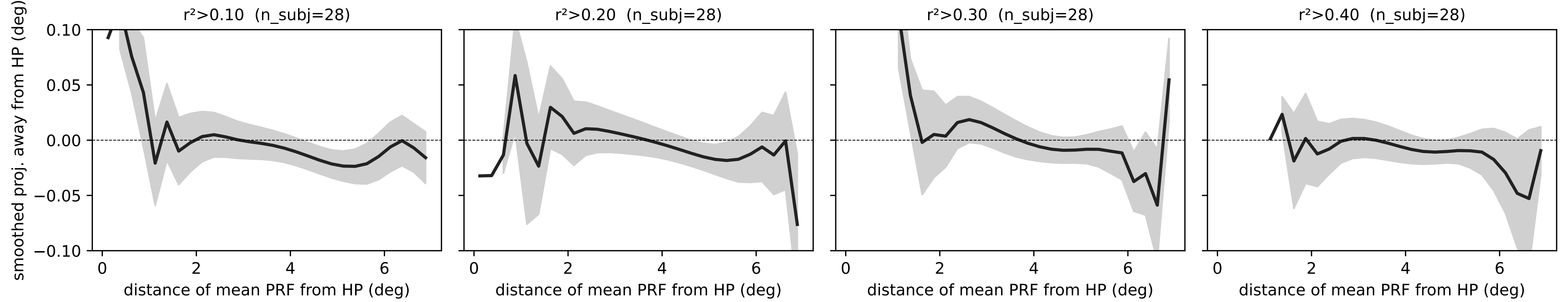

V3AB — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=28)
r²>0.30 (n_subj=28)
r²>0.40 (n_subj=28)
p=0.153
p=0.226
p=0.319
smoothed proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6

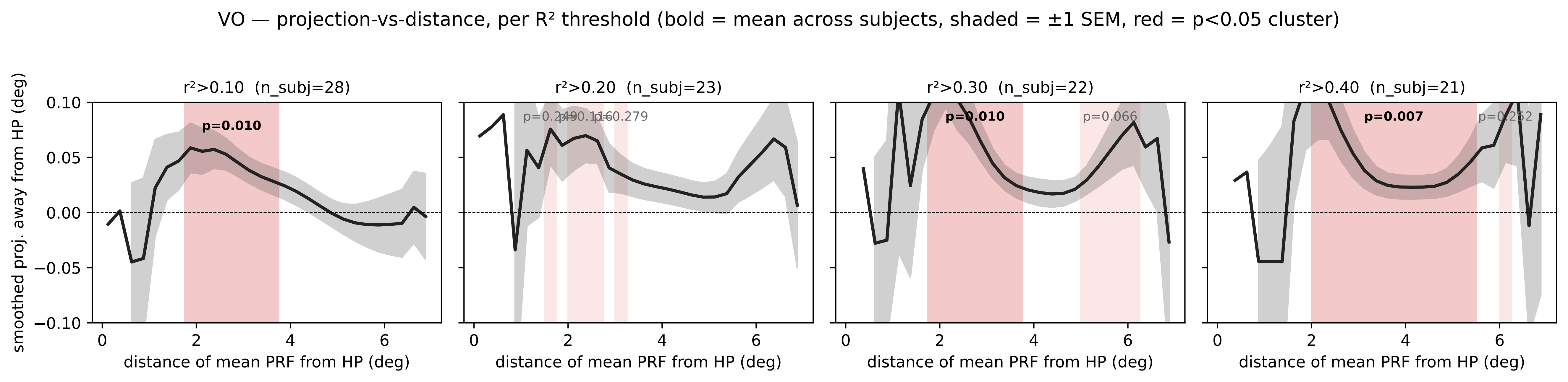


hV4 — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=28)
r²>0.30 (n_subj=28)
r²>0.40 (n_subj=28)
smoothed proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6


LO — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=28)
r²>0.30 (n_subj=28)
r²>0.40 (n_subj=28)
smoothed proj. away from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6
distance of mean PRF from HP (deg)

TO — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)





VO — projection-vs-distance, per R² threshold (bold = mean across subjects, shaded = ±1 SEM, red = p<0.05 cluster)
r²>0.10 (n_subj=28)
r²>0.20 (n_subj=23)
r²>0.30 (n_subj=22)
r²>0.40 (n_subj=21)
p=0.010
p=0.279
p=0.010
p=0.066
p=0.007
smoothed proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6